

JOSHUA KARTHIK ASHOK

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Backend & AI Systems Engineer with a track record of shipping AI-driven optimization platforms and scalable backend systems across enterprise and startup environments.

Education

Amrita Vishwa Vidyapeetham

B.Tech in Computer and Engineering

- CGPA: 7.59 (Expected date of graduation: April 2027)

Aug 2023 – Present

Coimbatore, TN

Work Experience

Hindustan Petroleum Corporation Limited (HPCL)

Project Intern – IMM Department

- Architected a **dual-solver vessel scheduling platform** (CP-SAT + MILP, **20K+ mixed-integer variables**, 200+ constraint templates) for HPCL's annual LPG import operations across **20+ ports**; reduced optimization time from **~10 hours to <1 hour** and achieved **89–97% demand accuracy** with dedicated Non-Designated port cost optimization.
- Built a **multi-pass LLM extraction engine** (Gemini 2.5 Flash + IBM Docling) that converts crude oil assay PDFs into validated Excel workbooks, automating the transcription of **200+ data fields per assay** backed by **11 physics-based validation checks** and a human-in-the-loop review dashboard.
- Deployed a **serverless executive dashboard** (Google Sheets → Apps Script API → auto-refreshing TV display) consolidating real-time data from **5 departments** into a single C-suite information screen, eliminating **45+ min/day** of manual cross-department reporting with traffic-light alerting and **60-second data refresh**.
- Automated daily consolidation of pipeline operations reports from **4 companies** (MPSPL, RRKPL, VVSPL, MDPL) by building a Python tool that fetches DOR attachments from Outlook, maps data blocks via **dynamic text anchors**, and safely injects values into a master tracking workbook — replacing daily manual copy-paste with a **scheduled, formula-safe, backup-protected sync**.

TruistyBytes

Full Stack Developer Intern

- Engineered **React frontends** and **Node.js/MongoDB APIs**, achieving **20–30% faster UI load times** and **25% improved API reliability** through optimized components, JWT auth hardening, and structured data validation.
- Strengthened product quality by adding **70% test coverage** for new features and maintaining CI pipelines, reducing deployment issues by **40%** while contributing to Agile sprint planning and code reviews.

Hyundai

SAP ABAP Intern

- Engineered and enhanced **SAP ABAP artifacts** — including **SmartForms**, **DDIC objects**, **BAPIs**, **BADIs**, and **user-exits** — streamlining core manufacturing and procurement workflows and enabling **30% reduction in manual processing overhead** across SAP modules (MM, SD).
- Optimized legacy ABAP programs through **performance tuning**, **SQL trace analysis**, and **buffering strategies**, improving **runtime efficiency by 20–25%** and elevating system reliability in production-grade environments.

Projects

ClarityLegal

GitHub 

AI-Driven Legal Document Intelligence Platform

- Engineered an AI-powered legal analysis system using **Vertex AI (Gemini)**, **OCR**, **RAG search**, **FastAPI**, and **Next.js**, enabling **<5 s document processing** and structured extraction of summaries, risks, and key clauses.
- Developed **clause extraction**, **risk scoring**, and **grounded Q&A** modules with near-zero hallucination; implemented semantic search over multi-document embeddings for real-time cross-document retrieval. Validated on SaaS, NDA, and employment contracts as a commercial prototype for SMEs.

RouteX

GitHub 

AI-Powered Maritime Fleet Optimization Platform

- Built an AI-driven coastal fleet optimization system using **Google OR-Tools (CP-SAT)**, **FastAPI**, **Next.js**, and **MongoDB**, solving HPCL's 9-tanker, 17-port problem with **~726 feasible route patterns** and **40–50% cost reduction** over manual planning.
- Developed a **constraint programming engine** with multi-objective support (cost, time, emissions), solver profiles (15 s–300 s), and parallel route generation; enforced single-port loading and max two-port discharge with **100% demand satisfaction**.
- Implemented real-time maritime visualization with **interactive route playback**, **Gantt-based fleet scheduling**, and **Celery/Redis** async processing; integrated **searoute-py** for sea distances. **Top 34 in poWer Lab 2.0 – HPCL Challenge 7.1**.

OptiWare

Live /GitHub 

Edge-AI Warehouse Automation

- Developed an edge-AI warehouse automation system using **Sony IMX-500**, **Raspberry Pi**, and **TensorFlow Lite**, achieving **sub-200 ms inference latency** with high-precision item detection for real-time sorting.
- Integrated **PostgreSQL** and **AWS (SNS/S3/Lambda)** to enable **<1.5 s event propagation**, automated alerting, and operator dashboards; optimized edge pipeline to reduce overhead by **30%**.

Publications

Navigating the Dual Realities of Image Generation and Deepfake Detection

Research Paper 

- Conducted a comparative study of GAN- and diffusion-based image generation models, identified key robustness gaps in deepfake detection (adversarial vulnerability, compression artefacts, cross-model inconsistency), and proposed methodological directions—spanning multimodal cues and feature-level analysis—to improve forensic reliability.

Achievements & Certifications

HP poWer Lab 2.0 (HPCL) — Top 34 / 40,000+ Semi-Finalist (Project: RouteX)

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SEED Business Challenge 2026 (University of Maryland) — Winner

LinkedIn 

Google Gen AI Exchange Hackathon — Top 100 / 30,000+ (Project: ClarityLegal)

LinkedIn 

Sony AITRIOS Hackathon — Top 10 / 10,000+ (Project: OptiWare)

LinkedIn 

Odoo Hackathon — Top 10 / 1,500 (Full-stack rental system)

Google Drive 

SIH 2024 — Selected to Regionals (Project: modernERP)

LinkedIn 

IITM Foundational Degree — Foundational in Data Science

View Certificate 

Technical Skills

Languages: Python, TypeScript/JavaScript, C/C++, Java, GoLang, SAP ABAP

Backend: FastAPI, Node.js (Express), Next.js API Routes

Frontend: React, Angular

Databases: PostgreSQL, MongoDB, Firestore, Redis, JPA2, Hibernate (ORM)

AI/ML & CV: OpenCV, TensorFlow, TensorFlow Lite, OR-Tools (CP-SAT), OCR pipelines, searoute-py

Cloud/DevOps: Docker, Git, CI/CD, Vercel, AWS (SNS/S3/Lambda), GCP Cloud Run, Celery/Redis